Paper Explainer Package

Comprehensive Technical Companion to:

Gradient-Free Stochastic Optimization of Navier-Stokes Active Contours for Unsupervised High-Throughput Phenotyping

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Auto-generated companion document — 2026-03-13

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1 Overview and Purpose

This document is a **structured explanatory companion** to the manuscript "Gradient-Free Stochastic Optimization of Navier-Stokes Active Contours for Unsupervised High-Throughput Phenotyping" by Ndubuisi and Kazic. It is **not** a new research paper. Instead, it serves as:

- A guided walkthrough of the manuscript from introduction to conclusion
- An audit of every experiment output on disk, with file paths
- An explanation of the mathematics, methodology, and literature
- A figure-by-figure and table-by-table interpretation guide
- A provenance record documenting which outputs are final, partial, or diagnostic

Manuscript location. `Paper_3/manuscript_final_strengthened/tex/Latest_Paper_1_strengthened.tex`
Compiled PDF: `Paper_3/manuscript_final_strengthened/pdf/Latest_Paper_1_strengthened.pdf`

2 What the Paper Studies

The paper addresses the problem of **unsupervised segmentation of plant disease lesions** in high-resolution 16-bit leaf images. Lesions exhibit enormous morphological diversity: sharp necrotic spots, faint chlorotic halos, coalescing regions, and complex backgrounds (venation, glare, dust).

The core insight is to treat the tuning of a **physics-based segmentation engine** (Navier-Stokes-inspired diffusion + active contour snakes) as a **per-image black-box optimization problem**, rather than fixing one parameter set for all images.

2.1 Why This Matters

- Deep learning requires thousands of pixel-accurate masks — unavailable in most phenotyping labs.
- Fixed-parameter physics models fail on heterogeneous datasets.
- Per-image optimization overcomes both limitations: no labels needed, and parameters adapt to each leaf.
- The resulting pseudo-labels can train a fast student network for deployment.

3 Core Scientific Objective

Develop a fully automated, unsupervised pipeline that:

1. Uses PDE-based diffusion to create smooth energy landscapes from image gradients
2. Evolves active contours (snakes) to segment lesions
3. Optimizes 7 physics parameters per image via gradient-free stochastic search
4. Scores segmentation quality without ground truth (heuristic objective)
5. Produces 16-bit masks and metadata for downstream phenotyping
6. Distills the optimizer's outputs into a fast UNet student

4 High-Level Method Summary

1. **Pre-processing**: 16-bit TIFF → 8-bit working space; channel fusion (green + grayscale + Lab b^*); CLAHE contrast enhancement; leaf masking.
2. **Energy field**: Navier-Stokes-inspired diffusion-reaction PDE smooths image gradients into a clean energy map (80 iterations, CFL-stable).
3. **Contour evolution**: Threshold energy map → candidate blobs → fit snakes with tension α, rigidity β, balloon γ.

4. **Scoring**: Heuristic objective combining gradient alignment, Lab color separability, topology reward, and penalty terms.
5. **Optimization**: Random-restart hill climbing (64 random + 48 refine evaluations per image) over bounded 7D parameter space.
6. **Output**: Per-image optimized masks, overlays, parameters, and scores.
7. **Distillation**: Patchify 174 leaves → 4,025 training patches → train UNet student (40 epochs, BCE+Dice loss).

5 Section-by-Section Explanation of the Manuscript

5.1 Section 1: Introduction

Motivates the work by identifying the gap between deep learning's data hunger and physics-based models' parameter sensitivity. Lists four contributions: (1) diffusion engine, (2) no-GT objective, (3) stochastic optimizer, (4) phenotyping-ready pipeline.

Note: Contains a TODO for biological experiment context (genotype, pathogen, DPI, camera) — this should be filled before submission.

5.2 Section 2: Related Work

Five subsections covering: active contours and level sets (Kass, Caselles, Chan-Vese, GVF), PDEs in vision (Navier-Stokes inpainting, PlantCV), modern segmentation (U-Net through SAM), hyperparameter optimization, and plant phenotyping tooling.

The literature positions the paper at the intersection of classical physics-based segmentation and modern automated parameter search, with phenotyping as the application domain.

5.3 Section 3: Problem Setup and Notation

Defines the image I, parameter vector $\theta = (\mu, \lambda, \alpha, \beta, \gamma, T)$, and the per-image optimization problem $\theta^* = \arg\max S(I; \theta)$.

Introduces two key mathematical concepts:

- **Candidate trace sets** $\mathcal{T}_i$ (Eq. 2): logging all evaluated candidates during optimization
- **Projected pairwise surfaces** $\mathcal{S}_{p,q}$ (Eq. 3): projecting the 7D landscape onto 2D parameter planes
- **Parameter-cloud geometry** (Section 3.2): PCA of global/local solutions, centroid shift, k-NN analysis

5.4 Section 4: Mathematical Framework

5.4.1 Navier-Stokes Diffusion (Eq. 4–5)

The diffusion-reaction PDE evolves an energy field $E^{(t)}$:

$$E^{(t+1)} = E^{(t)} + \Delta t\Big(\mu \Delta E^{(t)} + (\lambda + \mu)\,\mathrm{div}(\nabla E^{(t)}) + q \cdot (F - E^{(t)})\Big)$$

- μ: viscosity (Laplacian smoothing)
- $(\lambda + \mu)\,\mathrm{div}(\nabla E)$: elastic coupling (coherent structure)
- $F = \|(I_x - S_x, I_y - S_y)\|_2$: forcing from seed-vs-image gradient discrepancy
- $q = \mathbf{1}[G > T]$: edge-gating mask
- Stability: CFL condition $\Delta t \leq h^2/(4\mu)$; with $h = 1$, requires $\mu \leq 0.25$
- 80 iterations, normalize E to $[0, 255]$

5.4.2 Active Contour Dynamics (Eq. 6–7)

Snake energy functional with internal terms (α tension, β rigidity) and external energy from the diffused field. Balloon force γ drives outward expansion. Multi-lesion handling via energy-map thresholding with scale constraints ($f_{\min} = 5 \times 10^{-4}$, $f_{\max} = 0.45$).

5.5 Section 5: Methodology — Stochastic Parameter Search

5.5.1 Heuristic Objective (Eq. 9)

$$S(\theta) = w_g\, \mathcal{G}(M, I) + w_c\, \mathcal{C}(M, I) + w_n \log(1 + N(M)) - \mathcal{P}_{\text{area}}(M) - \mathcal{P}_{\text{noise}}(M)$$

- $\mathcal{G}$: Gradient alignment — boundaries on stronger-than-average gradients (Eq. 10)
- $\mathcal{C}$: Chromatic separability — Lab a^*b^* distance inside vs outside (Eq. 11)
- $\log(1 + N)$: Topology reward — multiple lesions with diminishing returns
- $\mathcal{P}_{\text{area}}$: Penalty for empty or too-large masks
- $\mathcal{P}_{\text{noise}}$: Penalty for ultra-small dust components

5.5.2 Search Algorithm (Algorithm 1)

Two-phase: (1) $K = 64$ uniform random samples, (2) top-4 seeds refined with $R = 12$ perturbation steps each, cooling $\sigma_j = \sigma_0 \cdot 0.85^j$. Total $\sim$112 evaluations per image.

5.6 Section 6: Implementation Details

16-bit handling, channel fusion, CLAHE, scale-aware filtering, downscaling during search ($s = 0.75$), timeouts, parallelism.

5.7 Section 7: Experiments

Dataset: 174 16-bit TIFF leaf images. Evaluation: no-GT (all), sparse GT (60 planned), cross-condition. UNet student: 4,025 patches, 40 epochs.

5.8 Section 8: Results

Detailed in Section 6 below.

5.9 Section 9: Multi-Stage Inference

Two-stage nested lesion approach: outer (green) + inner (magenta) masks. Algorithm 2.

5.10 Section 10: Discussion

Parameter transfer, morphology fidelity, landscape geometry, manifold structure, limitations.

5.11 Section 11: Conclusion

Summarizes contributions and points to amortized prediction as future work.

5.12 Appendices A–C

Additional optimizer diagnostics, supplementary trace plots, supplementary geometry plots.

6 Experiment-by-Experiment Explanation

6.1 Experiment 1: Production Optimizer Run (174 Leaves)

What	Per-image stochastic optimization on all 174 leaves
Why	Generate optimized masks and parameters for every image
How	Random-restart hill climbing (64 random + 48 refine)
Output	`opt_summary_local.csv`
Key result	100% OK; mean score 21.10; median runtime 242s
Paper role	Main paper — Figs 1–3, referenced throughout
Status	FINAL

6.2 Experiment 2: Objective Function Ablations

What	Remove one objective term at a time
Why	Justify each term in the composite objective (Eq. 9)
How	Same optimizer, modified weights, 30-image stratified subset
Output	`ablations_objective/` (6 subdirectories)
Key result	Contour-count removal: score 14.80 $\rightarrow$ 3.68 (most impactful)
	Noise penalty removal: n_c 134 $\rightarrow$ 188 (fragmentation)
Paper role	Main paper — Table I
Status	FINAL

6.3 Experiment 3: Search Strategy Ablations

What	Modify one search knob at a time
Why	Justify the two-phase search design
How	Same objective, modified search params, 30-image subset
Output	`ablations_search/` (5 subdirectories)
Key result	No-downscale highest score (16.58) but 34% slower
	Random-only competitive (14.88 vs 14.95)
Paper role	Main paper — Table II
Status	FINAL

6.4 Experiment 4: Teacher–Student Distillation

What	Train UNet student on optimizer pseudo-labels
Why	Amortize per-image search into fast inference
How	4,025 patches, 40 epochs, BCE+Dice, threshold 0.30
Output	`teacher_student_metrics.csv`,
	`teacher_student_metrics_roi_thr0p30.csv`,
	`teacher_student_figures/`, `threshold_sweep.csv`
Key result	Baseline Dice 0.598 (all), 0.642 (teacher-nonempty)
	18 failures, 12 teacher-empty disagreements
Paper role	Main paper — Section 8.1, Figs 4–6
Status	FINAL

6.5 Experiment 5: 9-Leaf Representative Trace Study

What	Instrumented rerun with candidate-level trace logging
Why	Characterize objective landscape geometry
How	9 selected leaves (3 high/3 median/3 low score); identical settings as production; SLURM job 12783426
Output	`objective_surface_traces_representative_fullbudget_9leaf/`
Provenance	ALL traces from single dedicated rerun — NO fallback
Key result	High-score: sharp landscapes, refine gain +5.77 Low-score: flat landscapes, refine gain +0.56 Dominant pair: $T \times \beta$ (7/9 cases)
Paper role	Main paper — Figs 9–12, Section 8.4
Status	FINAL — homogeneous dedicated rerun

6.6 Experiment 6: Parameter-Cloud Manifold Geometry

What	PCA/t-SNE/UMAP analysis of 7D parameter space
Why	Characterize structure of optimizer's solution space
How	150 global/local parameter pairs from 15 leaves
Output	`parameter_shift_manifold_geometry/`
Key result	2 PCs capture 46.5% variance; k-NN 0.534 (vs 0.33 random)
Paper role	Main paper — Figs 13–14, Section 8.5
Status	FINAL

6.7 Experiment 7: Solid-3D Parameter Geometry

What	Covariance ellipsoids, convex hulls, density isosurfaces
Why	Volumetric characterization of parameter clouds
How	2-sigma ellipsoids in PCA space; hull analysis
Output	`parameter_shift_3d_analysis_solid/`
Key result	Centroid shift 1.616 std units; partial ΔD separation
Paper role	Main paper — Fig 15; Appendix C
Status	FINAL

6.8 Experiment 8: Patch Morphology Transfer

What	Test whether per-leaf params transfer to 512×512 patches
Why	Validate global-to-local parameter stability
How	150 patches from 15 leaves; global vs local re-optimization
Output	`patch_local_opt_v2.csv`, `patch_transfer_morphology_eval.csv`, `patch_transfer_morphology_plots/`
Key result	Median Dice 0.78; area underestimated (0.026 vs 0.169); local re-opt reduces count error 35%
Paper role	Main paper — Table III, Figs 17–19, Discussion
Status	FINAL

7 Figure-by-Figure Explanation with File Paths

7.1 Main Text Figures

7.1.1 Fig 1 — Optimizer Status Bar

`optimizer_summary/status_bar.png`
Shows bar chart of optimizer run status for all 174 images. All bars show "OK" status.
Takeaway: 100% convergence proves pipeline reliability across diverse phenotypes.

7.1.2 Fig 2 — Runtime Histogram

`optimizer_summary/elapsed_seconds_hist.png`
Histogram of per-image wall-clock time (seconds). Median ∼250s.
Takeaway: Per-image optimization takes ∼4 minutes on CPU — practical for 174 images.

7.1.3 Fig 3 — Score Histogram

`optimizer_summary/score_hist.png`
Distribution of objective scores across 174 images. Range 11–46, mean 21.10.
Takeaway: Scores are unimodal with no catastrophic failures.

7.1.4 Fig 4 — Threshold Sweep

`threshold_sweep_plot.png`
Dice as a function of probability threshold on stitched UNet outputs.
Takeaway: Best at 0.3; teacher-empty disagreements persist regardless of threshold.

7.1.5 Figs 5–6 — Teacher-Student Montages

`teacher_student_figures/montage_teacher_student.png`
`paper_disagreement_montage_roi_thr0p30.png`
Fig 5: worst/median/best comparisons. Fig 6: teacher-empty disagreement cases.
Takeaway: Student approximates teacher well; failures are semantic (inner-mask definition), not background leakage.

7.1.6 Figs 7–8 — Parameter-Morphology Correlations

`optimizer_summary/ (derived scatter plots)`
Fig 7: Energy threshold vs. lesion count (inverse). Fig 8: Balloon force vs. area fraction (positive).
Takeaway: Optimized parameters have physically interpretable relationships with output morphology.

7.1.7 Figs 9–11 — Trace-Based Objective Surfaces

`objective_surface_traces_representative_fullbudget_9leaf/surface_plots/`
Projected T–β surfaces for high (DSC_0198), median (DSC_0379), and low (DSC_0261) score leaves.
Takeaway: Landscape curvature decreases with phenotypic difficulty. High-score leaves have sharp optima; low-score leaves are flat.

7.1.8 Fig 12 — Convergence Trajectory

`objective_surface_traces_representative_fullbudget_9leaf/trajectory_plots/`
Best-score-so-far vs. candidate index for DSC_0198 (high-score).
Blue = random phase, orange = refine phase.
Takeaway: Clear phase transition; refinement adds measurable gain on favorable landscapes.

7.1.9 Fig 13 — PCA 2D Global-Local Projection

`parameter_shift_manifold_geometry/plots/pca2d_global_local_dD.png`
Global (circles) and local (triangles) parameters in PCA space, colored by ΔD.
Takeaway: PC1 separates the two clouds; centroid shift magnitude 1.616 standardized units.

7.1.10 Fig 14 — Mean Parameter Shift Profile

`parameter_shift_manifold_geometry/plots/mean_delta_parameters.png`
Bar chart of mean shift per parameter.
Takeaway: Energy threshold dominates — local re-optimization primarily adjusts edge sensitivity.

7.1.11 Fig 15 — Covariance Ellipsoids

`parameter_shift_3d_analysis_solid/pca3d_covariance_ellipsoids.png`
2-sigma ellipsoids for global (blue) and local (red) clouds in PCA space.
Takeaway: Overlapping but displaced ellipsoids confirm structured parameter organization.

7.1.12 Fig 16 — Qualitative Grid

`teacher_student_figures/ (qual_grid)`
6-row grid from Dice≈0 to Dice≈0.84. Columns: input, teacher overlay, student overlay, masks.
Takeaway: Visual evidence spanning the full quality spectrum.

7.1.13 Figs 17–19 — Morphology Transfer Plots

`patch_transfer_morphology_plots/hist_area_frac_error.png`
`patch_transfer_morphology_plots/hist_count_error.png`
`patch_transfer_morphology_plots/scatter_teacher_vs_pred_area_frac.png`
Area error histogram, count error histogram, teacher vs. predicted scatter.
Takeaway: Both parameter sets underestimate area; local re-opt reduces count error 35%.

7.2 Appendix Figures

7.2.1 Appendix A — Optimizer Output Diagnostics

`optimizer_summary/n_contours_hist.png` and `optimizer_summary/area_px_hist.png`
Connected-component count and mask area distributions. Seven parameter histograms (one per parameter).

7.2.2 Appendix B — Additional Trace Trajectories

`objective_surface_traces_representative_fullbudget_9leaf/trajectory_plots/`
Six additional convergence trajectories (2 per tier: DSC_0166+0163, DSC_0274+0205, DSC_0185+0199).

7.2.3 Appendix C — Supplementary Geometry

`parameter_shift_3d_analysis_solid/pca3d_convex_hull_global_local.png`
`parameter_shift_3d_analysis_solid/pca3d_positive_vs_negative_dD_surfaces.png`
`parameter_shift_manifold_geometry/plots/tsne2d_delta_ddgroup.png`
`parameter_shift_manifold_geometry/plots/pca_centroid_shift_groups.png`
Convex hulls, ΔD surfaces, t-SNE, centroid shift by group.

8 Table-by-Table Explanation with File Paths

8.1 Table I — Objective Ablations

Source: `ablations_objective/` (6 subdirectories)
Section: Results 8.3
Format: 6 rows (full + 5 ablated variants) × 6 columns (OK%, Non-empty%, Med. score, Med. area, Med. n_c, Med. time)

How to read: Compare each ablated row against the "Full objective (ours)" row. Arrows indicate preferred direction (↑ or ↓).

Key findings:

- Removing contour count ($w_{\text{topo}} = 0$): score drops from 14.80 to 3.68 — **most impactful term**
- Removing noise penalty ($w_{\text{small}} = 0$): n_c jumps from 134 to 188 — fragmentation
- Removing gradient alignment ($w_{\text{grad}} = 0$): score drops to 9.71
- Removing color distance or area penalty: modest effects

8.2 Table II — Search Strategy Ablations

Source: `ablations_search/` (5 subdirectories)
Section: Results 8.3
Format: 5 rows × 6 columns (same as Table I)

Key findings:

- No-downscale: highest score (16.58) but 34% more runtime (5427.9s vs 4046.5s)
- Random-only: surprisingly competitive (14.88 vs 14.95)
- Low budget ($K = 16$): still achieves 14.91 — diminishing returns from more random samples
- Fast timeout (60s): modest score reduction (14.44)

8.3 Table III — Patch Morphology Transfer

Source: `patch_transfer_morphology_eval.csv` + `patch_local_opt_v2.csv`
Section: Discussion
Format: Metrics for All/Interior/Boundary patches, comparing global vs. local parameters against teacher

Key findings:

- Dice is modest: global 0.430, local 0.443 (+0.013)
- Area fraction severely underestimated: 0.026 vs. teacher 0.169
- Count error reduced 35% by local re-optimization: 19.1 → 12.4
- Interior patches benefit more from local re-opt than boundary patches
- **Conclusion**: Dice alone is insufficient for phenotyping quality; morphological metrics matter

9 Literature and Mathematics Explained

9.1 Literature Landscape

The paper draws on five distinct literature threads:

1. **Active contours**: Kass et al. (1988) → Caselles (1997, geodesic) → Chan-Vese (2001, region-based) → GVF (Xu & Prince, 1998). The paper extends this tradition by adding automated per-image parameter search.
2. **PDEs in vision**: Bertalmio et al. (2001, Navier-Stokes inpainting) inspires the diffusion formulation. The paper's energy-field diffusion is a scalar version of GVF with fluid-dynamics-inspired elastic coupling.
3. **Modern segmentation**: U-Net (2015) through SAM (2023). Used as context and baselines. The student model is U-Net-based.
4. **Black-box optimization**: Yu & Zhu (2020) survey; Hansen & Ostermeier (2001, CMA-ES); Jones et al. (1998, EGO). Justifies gradient-free approach for non-differentiable pipelines.
5. **Plant phenotyping**: Gehan & Kellogg (2017), Mahlein (2016), PlantCV (2025), Singh et al. (2016). Establishes the application domain and emphasizes trait-producing pipelines.

9.2 Mathematical Components

11 numbered equations in the manuscript:

- Eq. 1, 8: Per-image optimization formulation
- Eq. 2: Candidate trace set definition
- Eq. 3: Projected pairwise surface definition
- Eq. 4: Diffusion-reaction PDE update
- Eq. 5: Forcing term from gradient discrepancy
- Eq. 6: Snake energy functional
- Eq. 7: Snake evolution equation
- Eq. 9: Composite heuristic objective
- Eq. 10: Gradient alignment term
- Eq. 11: Chromatic separability term

Plus the necrosis ratio formula in Section 9 and the Dice/overlap formulations in the evaluation.

10 Main Findings and What They Mean

1. **100% convergence across 174 diverse leaves**: The pipeline is robust and does not fail catastrophically on any image.

2. **Ablations justify every component**: Contour-count topology reward is the most impactful objective term; the full search pipeline balances quality and speed.

3. **Teacher-student distillation works** (median Dice 0.682 on teacher-nonempty images), but 12 teacher-empty disagreements indicate a semantic gap between inner-mask and outer-lesion definitions.

4. **Score-tier-dependent landscape geometry**: The objective landscape is sharp for morphologically favorable images (high scores) and flat for challenging ones (low scores). This has practical implications for adaptive budget allocation.

5. **Structured parameter manifold**: 46.5% of variance in 2 PCs; shift direction partially predicts optimization success (k-NN 0.534 vs. 0.33 random). This supports amortized parameter prediction.

6. **Morphological fidelity diverges from overlap metrics**: Dice alone underestimates issues; area fraction is severely underestimated under patchification (0.026 vs. 0.169). Local re-optimization helps with count (35% error reduction) but not area.

11 Strongest Evidence

1. **Production run** (`opt_summary_local.csv`): 100% convergence on 174 heterogeneous leaves
2. **Ablation tables** (Tables I–II): Each component rigorously justified on 30-image subset
3. **9-leaf trace surfaces** (Figs 9–12): Clean provenance (single SLURM job); score-tier pattern
4. **Manifold geometry** (Figs 13–15): Quantified structure with k-NN above chance
5. **Morphology transfer** (Table III): Reveals Dice-morphology divergence important for phenotyping

12 What is Provisional, Appendix-Only, or Diagnostic

12.1 Provisional / Incomplete

- `objective_surface_traces_representative_fullbudget_plus/`: manifold and solid-3D subdirs EMPTY
- `param_predictor/`: EMPTY — future work (amortized prediction)
- Biological experiment context TODO in manuscript (genotype, pathogen, DPI)
- Sparse GT evaluation on 60-image subset: protocol described but annotation not completed
- Baseline comparisons (Otsu, labeled U-Net, SAM): recommended but not reported

12.2 Appendix-Only

- Parameter histograms (Fig A2): useful for reproducibility, not primary evidence
- Additional convergence trajectories (Appendix B): extend Fig 12 pattern
- Supplementary geometry (Appendix C): t-SNE, hulls, centroid groups
- Full-population traces (`objective_surface_traces_fullstudy/`): population evidence

12.3 Diagnostic Only — Not for Paper

- Codex-style variants (`optimization_surface_analysis_codex/`, `parameter_shift_3d_analysis_solid_codex`
- Per-leaf montage PNGs (`montage_patch_opt_v2_*.png`)
- `postprocess_sweep.csv`: parameter sweep diagnostic
- `optimizer_full/`, `optimizer_full_alt/`: empty directories
- `experiment_summary_master/`: prior summary attempt

13 Manifold / Solid-3D / Optimization Geometry Assessment

Question: Are manifold and solid-3D geometry figures already present?

Answer: **Yes** — all geometry figures are already present in the outputs and are included in the current manuscript.

- **PCA manifold figures**: Present in `parameter_shift_manifold_geometry/plots/` (30 files) and copied to `figs_from_outputs/manifold/` (6 files used in paper)
- **Solid-3D figures**: Present in `parameter_shift_3d_analysis_solid/` (25 files) and copied to `figs_from_outputs/solid3d/` (4 files used in paper)
- **Merged final figures**: Present in `optimization_manifold_and_solid3d_final/figures/` (19 files)

- **9-leaf trace geometry**: Present in `objective_surface_traces_representative_fullbudget_9leaf/` and copied to `figs_from_outputs/9leaf/` (54 files)

No new geometry figures needed. All required manifold, PCA, UMAP, t-SNE, solid-3D hull, ellipsoid, and density plots already exist on disk and are reflected in the manuscript.

14 Directory Structure Summary

Key paths (all relative to `experiments/exp_v1/outputs/`):

```
experiments/exp_v1/outputs/
  opt_summary_local.csv                          <- CORE: 174-leaf results
  ablations_objective/                           <- Table I source
  ablations_search/                              <- Table II source
  objective_surface_traces_representative_fullbudget_9leaf/
    candidate_traces/                            <- 9 trace CSVs
    surface_plots/                               <- 27+ surface plots
    trajectory_plots/                            <- 54+ trajectory plots
    reports/                                     <- surface_report.pdf/.txt
  parameter_shift_manifold_geometry/
    plots/                                       <- 30 manifold plots
    reports/                                     <- 3 reports
    data/                                        <- 4 CSVs
  parameter_shift_3d_analysis_solid/             <- 25 solid-3D files
  optimization_manifold_and_solid3d_final/       <- 19 merged figures
  optimizer_summary/                             <- 17 summary files
  teacher_student_figures/                       <- montages + plots
  patch_transfer_morphology_plots/               <- 7 morph plots
  teacher_student_metrics.csv                    <- baseline metrics
  teacher_student_metrics_roi_thr0p30.csv        <- ROI-gated metrics
  threshold_sweep.csv + threshold_sweep_plot.png <- sweep data + plot
  failures_dice_lt_0p3.csv                       <- 18 failure cases
  teacher_empty_cases.csv                        <- 12 disagreements
  patch_local_opt_v2.csv                         <- 150 patch results
  patch_transfer_morphology_eval.csv             <- morphology metrics
  patches/                                       <- training data
```

15 Final Summary

This paper presents a novel approach to unsupervised lesion segmentation that bridges physics-based modeling and automated parameter optimization. The key innovation is treating per-image physics parameter tuning as a black-box optimization problem, enabling robust segmentation across 174 morphologically diverse leaf images without any ground-truth labels.

The experimental evidence is comprehensive:

- Production run achieves 100% convergence (strong reliability)
- Ablations justify every objective and search component
- Trace-based landscape analysis reveals score-tier-dependent geometry
- Manifold analysis confirms structured, low-dimensional parameter organization
- Teacher-student distillation enables fast inference (with known semantic limitations)
- Morphology transfer analysis reveals important Dice-morphology divergence

The main gaps are: (1) no quantitative baseline comparisons, (2) sparse GT annotation not yet completed, (3) biological context TODO unfilled, (4) amortized parameter predictor not yet built.

All experiment outputs have clean provenance. The 9-leaf trace study (the most novel analysis) uses a single homogeneous SLURM rerun with no fallback or reused traces. No partial or mixed-provenance results are presented as final in the main paper.